

End of game

Radical, the developers of the game Prototype lay off half its staff

Project Natal

Prototype pictures finally make their way to the internet

Workshop

Gaming Handsfree

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Now we're all big fans of gaming and while futuristic hand gloves aren't going to hit us anytime soon, motion tracking is a nice thing to have in games – you can move your character around by using your hands for example. Project Natal seems to be shaping up well, but it still isn't here. With just a web camera, you can now have a motion tracking device for free. Use software called Free Track (www.free-track.net) and you can soon control your game views using your head.

The plot is simple – the web camera detects a set number of points on your head. One of the things that greatly improve the tracking performance is use of LEDs. These are things you can easily buy off the market and build yourself. A few PDFs are available with guides on how to build your own head gear.

Choose point model

The first step is to set the point model that you will be using with Free Track. The larger point model you use, the better. It allows for better detection as well as it tracks movements along various axes. Click on the Model button at the top of the Free Track window. Click on the point model that you wish to use. The single point model should do just fine if you wish to control the game in two degrees. This is perfect for racing games for example. It will let you move the player

view to either side to see if you're being overtaken. Flight simulators will also benefit. If you select any of the other point models, you need to specify other values for the distance between the two points.



Selecting the point model to be used by Free Track

Setting up the cam

Make sure that the web camera is mounted exactly in front of you and there are no bright lights in the background. Next, configure the camera in a way that it detects the points easily and doesn't lose track of them. LEDs are simpler to detect as compared to bright spots

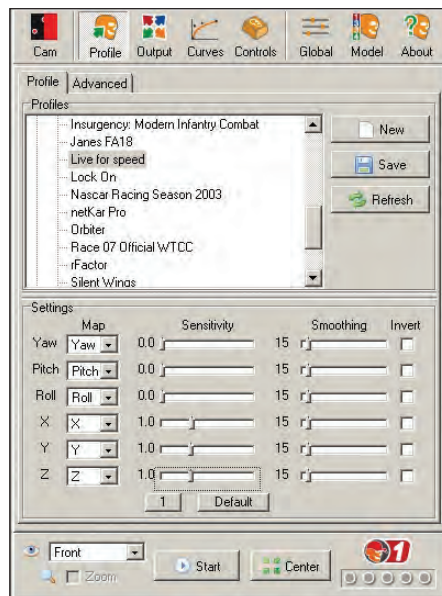
marked on a cap or a head- phone. Click the Cam button and then on the Start button after selecting your camera from the dropdown menu. Next, set the threshold slider till Free Track has a clear lock on the guide points. The background should go blank and the LEDs should stand out in it. Move your head around to ensure Free Track locks on the points properly. If it doesn't, you'll have lots of problems with controlling during game play.

There are also some other controls you can change to tweak and improve the motion detection accuracy. The point size menu allows you to specify the size of the LED or markings. You can specify the approximate size of the LEDs or markings and if possible, the frame rate of the camera.

Sensitivity

Once you have the detection set up fine, it's time to fine tune Free Track to our requirements. Every game requires a different setting – remember that you can use Free Track as a tool to move the player view or even to control the player itself or look at an object from different directions. Profiles can be customised for every game.

Click on Profile, and then select one of the games from the list. Change the sensitivity and smoothing levels from this menu and watch the live preview of the controls on the right.



Customising game profiles and setting sensitivities

The 3D model should resemble your head movements. Further tweaking of movement and rotation is possible by clicking on the Curves button on the toolbar. New profiles can also be created for games that aren't natively supported.

Enabling input devices

Free Track can be setup in a way that your head movements control the key strokes or even mouse movements. Click on Output, then on the Keyboard tab. Click the checkbox and then click on the button adjoining the different directions. Set a key that you want pressed when this movement is done. This way, you can even control applications on the desktop using. For example, you can load up Firefox when you move your head down. For flight simulations such as Microsoft Flight



Control key strokes using head movements

Simulator 2004 and 2002, you can use the FSUIPC add-on.

All that's left is to start the game, set your keys and profiles although Free Track's profiles should take care of everything. You're ready to go!